Rohit Roy

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About

I am a 2nd year PhD student at Centre Inria d'UniCA, working on problems in computational geometry. I previously developed a strong interest in probability theory, and my research focus has since shifted toward Topological Data Analysis and exploring topological solutions to data-driven problems.

Education

Centre Inria d'Université Côte d'Azur — *Nov 2025 – Oct 2028 (expected)*
PhD in Computer Science and Applied Mathematics

Chennai Mathematical Institute — *Aug 2023 – Apr 2025*
Master of Science in Data Science — CGPA : 9.00

Chennai Mathematical Institute — *Dec 2020 – Apr 2023*
Bachelor of Science (Honours) in Mathematics and Computer Science — CGPA : 7.88

Publications

Conference Proceedings

[1] S. Pritam, **R. Roy**. Collapse and Persistence of Directed Filtered Graphs. DOI
SIAM Symposium on Algorithm Engineering and Experiments (ALENEX26)

Workshop Papers

[2] E.W. Chambers, C. Fillmore, S.S. Mukherjee, **R. Roy**, E. Stephenson, M. Wintraecken. The symmetry set as a knitting pattern for vineyards.
Computational Geometry: Young Researchers Forum (CG:YRF) 2026

[3] E.W. Chambers, C. Fillmore, S.S. Mukherjee, **R. Roy**, E. Stephenson, M. Wintraecken. Bouquet: A Visualization Tool for Symmetry Sets and Vineyards. Tool, HAL
European Workshop on Computational Geometry (EuroCG 2026)

Preprints

[4] E.W. Chambers, C. Fillmore, S.S. Mukherjee, **R. Roy**, E. Stephenson, M. Wintraecken. The Singular Source of Vineyard Monodromy. arXiv, HAL

[5] S. Chowdhury, S. Pritam, **R. Roy**, M.C. Sajeev. Filtration-Based Representation Learning for Temporal Graphs. arXiv

Presentations

Talks

Monodromy in Radial Persistence Vineyards and visualizing with Bouquet2D — *Oct 2026*
DataShape Seminar, Inria Sophia-Antipolis Slides

Bouquet: A Visualization Tool for Symmetry Sets and Vineyards — *Mar 2026*
European Workshop on Computational Geometry (EuroCG 2026), Hagen Slides

Posters

Bouquet: A Visualization Tool for Symmetry Sets and Vineyards — *Sep 2026*
AATRN Poster Session Poster

Research Experience

Research Intern *Bordeaux, France*
LaBRI, Universite de Bordeaux *May 2023 – Jul 2023*

- Worked under Prof. Jean-Francois Markert on directed animals in 2D.
- Investigated open combinatorial questions concerning directed animals, a concept representing percolation clusters in directed percolation models and conducted simulations of directed animals on triangular and square lattices.
- More details can be found here, with the complete internship log here.

Reading Project: Branching Processes and Convergence Concepts *May 2022 – Jul 2022*
Under Prof. Ayan Bhattacharya, IIT Bombay

Reading Project: Markov Chains and related concepts *May 2022 – Jul 2022*
Under Prof. Rajeeva Karandikar, Chennai Mathematical Institute

Teaching Experience

Chennai Mathematical Institute *Chennai, India*
Teaching Assistant for the following courses

- Data Mining and Machine Learning *Jan 2025 – Apr 2025*
  Instructor: Prof. Madhavan Mukund · For: M.Sc. Data Science 1st Year
- Distributed Computing and Big Data *Jan 2025 – Apr 2025*
  Instructor: Prof. Venkatesh Vinayakrao · For: M.Sc. Data Science 1st Year
- Introduction to Martingales *Jan 2025 – Apr 2025*
  Instructor: Prof. Rajeeva Karandikar · For: B.Sc. 3rd Year and M.Sc.
- Natural Language Processing *Aug 2024 – Dec 2024*
  Instructor: Prof. Ramaseshan Ramachandran · For: M.Sc. Data Sci. & M.Sc. Comp. Sci.
- Probability Theory *Jan 2024 – Apr 2024*
  Instructor: Prof. Rajeeva Karandikar · For: B.Sc. Mathematics 1st Year
- Mathematical Methods in Analysis *Aug 2022 – Nov 2022*
  Instructor: Prof. Rajeeva Karandikar · For: M.Sc. Data Science 1st Year

Academic and Volunteer Service

- **Artifact Evaluation Committee member**, SIAM Symposium on Algorithm Engineering and Experiments (ALENEX 2027).

Awards

- **Olympiad Mathematics :** Qualified for the Indian National Mathematical Olympiad (INMO) in 2018.
- **KVPY-SA 2018 Fellow :** Recipient of the KVPY scholarship given yearly.
- **Mathematics Talent Reward Programme (MTRP)** held in ISI Kolkata : Qualified in 2017 and 2019, and received 3rd prize in 2017.

Industry Experience

Machine Learning Engineer *Bengaluru, India*
Hewlett-Packard, Printing and PC Systems (PPS) *Jun 2025 – Oct 2025*

- Contributed to the development of an enterprise-scale question–answering system for printing solutions at HP.
- Extended a document-based language model pipeline to support multimodal inputs, incorporating information from images and tabular data.
- Implemented and evaluated models and workflows on the AWS SageMaker platform.
- Investigated and applied prompt engineering techniques to improve model interpretability and response quality.
- Designed mechanisms for structured metadata retrieval and document access from SharePoint to support efficient information retrieval.

Data Analysis Intern (Industry Internship) *Chennai, India*
NPTEL, IIT Madras *May 2024 – Aug 2024*

- Classified colleges on their performance in NPTEL courses using clustering and came up with metrics for comparison.
- Created System Dynamic (SD) models using Vensim to mimic NPTEL course enrollment and registration trends across semesters with real world factors in play.

Languages

Bengali : Native **English :** C2 **Hindi :** C2 **French :** A1 **German :** A1